FINAL REPORT

Introduction

Cryptocurrency markets experience rapid price fluctuations due to market sentiment, trading activity, regulatory announcements, and investor behavior. Predicting volatility helps investors minimize risk and improve trading decisions.

This project develops a machine learning system to forecast cryptocurrency volatility using historical OHLC prices, trading volume, and market capitalization. The system applies preprocessing, feature engineering, and machine learning algorithms to generate accurate predictions.

Problem Statement Summary

The objective of this project is to predict future cryptocurrency volatility levels using machine learning techniques. Historical market data is analyzed to identify patterns and relationships that influence volatility.

Dataset Description

The dataset contains:

- Date
- Cryptocurrency Symbol
- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume
- Market Capitalization

The dataset includes records from multiple cryptocurrencies over different time periods.

Data Preprocessing

The following preprocessing steps were performed:

Missing Value Handling

- Null values replaced using forward fill or mean values

Data Cleaning

- Duplicate records removed
- Invalid rows filtered

Feature Scaling

- Standard Scaler applied

Exploratory Data Analysis (EDA)

Key Observations

1. Cryptocurrency prices show highly volatile behavior.
2. Trading volume is positively correlated with volatility.
3. Bitcoin and Ethereum exhibit relatively stable trends compared to smaller coins.
4. Sudden spikes in volume often lead to large price fluctuations.

Visualizations Used

- Line charts
- Correlation heatmaps
- Rolling volatility graphs

Feature Engineering

The following features were engineered:

Feature	Purpose
Rolling Mean	Identify trend
Rolling Std Dev	Measure volatility
Bollinger Bands	Detect market conditions

Liquidity Ratio	Volume/Market Cap

Model Development

Models Used

Model	Purpose
Linear Regression	Baseline model
Random Forest	Non-linear prediction
XGBoost	High-performance boosting model

Model Evaluation

Evaluation Metrics

Metric	Description
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
R^2 Score	Goodness of fit

Sample Results

Model	RMSE	MAE	R^2 Score
Linear Regression	0.156	0.052	0.007
Random Forest	0.147	0.031	0.109
XGBoost	0.144	0.039	0.146

Best Performing Model

XGBoost achieved the best performance due to their ability to capture non-linear patterns

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Challenges Faced

- High market volatility
- Missing and inconsistent data
- Feature selection complexity
- Overfitting in complex models

Future Improvements

- Real-time API integration
- Advanced deep learning models
- Sentiment analysis integration
- Real-time trading alerts
- Cloud deployment

Conclusion

This project successfully developed a machine learning pipeline for cryptocurrency volatility prediction. Multiple models were trained and evaluated using historical cryptocurrency data. XGBoost and LSTM models produced strong performance in forecasting volatility patterns.

The system can help traders, analysts, and financial institutions better understand market behavior and manage investment risks effectively.